

INERTIA OPERATOR (ALGEBRIC FORM)

6/

$$\begin{aligned} {}^0\mathbf{I}_B(\underline{m}) &= \int_V \left[{}^0\mathbf{z}_{P/B} \times \right] \left(\underline{m} \times \mathbf{z}_{P/B} \right) \rho dV = \\ &\quad \uparrow \\ &\quad - \left[{}^0\mathbf{z}_{P/B} \times \right] \underline{m} \\ &= \left(\int_V \underbrace{- \left[{}^0\mathbf{z}_{P/B} \times \right] \left[{}^0\mathbf{z}_{P/B} \times \right]}_{\left[{}^0\mathbf{r}_{P/B} \times \right]^T} \rho dV \right) \underline{m} = \end{aligned}$$

Then we obtain

$${}^0\mathbf{I}_B = \int_V \left[{}^0\mathbf{z}_{P/B} \times \right]^T \left[{}^0\mathbf{z}_{P/B} \times \right] \rho dV \in \mathbb{R}^{3 \times 3}$$

NOTE It is convenient to compute ${}^x\mathbf{I}_B$ w.r.t. a frame $\langle x \rangle$ fixed w.r.t. the body so that ${}^x\mathbf{I}_B$ is constant

$${}^0\mathbf{I}_B = {}^0\mathbf{R}^b \mathbf{I}_B {}^b\mathbf{R}^0$$

↑
constant

Let $\overset{b}{P}(P-B) = \overset{b}{Z}_{P/B} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ then
 body fixed

7/

$$\overset{b}{I}_B = \int_V \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}^T \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} \rho dV$$

$$\begin{bmatrix} 0 & z & -y \\ -z & 0 & x \\ y & -x & 0 \end{bmatrix} \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$$

symmetric $\rightarrow$

$$\begin{bmatrix} z^2 + y^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix}$$

Then

$$\overset{b}{I}_B = \begin{bmatrix} \int_V (z^2 + y^2) \rho dV & -\int_V xy \rho dV & -\int_V xz \rho dV \\ * & \int_V (x^2 + z^2) \rho dV & -\int_V yz \rho dV \\ * & * & \int_V (x^2 + y^2) \rho dV \end{bmatrix}$$

positive def.
 symmetric matrix (*)

$$\Rightarrow \underline{\underline{u}}^b \overset{b}{I}_B \underline{\underline{u}}^b > 0 \quad \forall \underline{\underline{u}}^b \neq 0$$

(*) Except non physical pathologic models

FACTS ABOUT SYMMETRIC MATRICES

8/

- $A = A^T$ has always real eigenvalues
- $A = A^T$ can always be diagonalized by orthonormal matrices i.e.

$$A = V \Sigma V^T$$

$$V^{-1} = V^T$$

$$\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_m \end{bmatrix}$$

eigenvalues of A

$$V = [\underline{v}_1 \dots \underline{v}_n]$$

eigenvectors
of A

- $A = A^T$ is positive/negative definite
(i.e. $\underline{x}^T A \underline{x} > 0$ / $\underline{x}^T A \underline{x} < 0 \quad \forall \underline{x} \neq 0$)
iff. $\sigma_i > 0$ / $\sigma_i < 0 \quad \forall i$
eigenvalues of A

Properties of inertia matrix

9/

- ${}^b I_B = {}^b I_B^T$ then $\exists \mathbb{R}^3$ (body fixed) s.t. ${}^a I_B$ is diagonal

Proof Since ${}^b I_B = {}^b I_B^T$ then $\exists V \in \mathbb{R}^{3 \times 3}$ s.t.

$${}^b I_B = V^T \Sigma V$$

V is orthonormal then it is a rotation matrix therefore $V^T \Sigma V$ can be seen as a coordinate transformation of operator Σ .

$$\Sigma = {}^a I_B = V {}^b I_B V^T = {}^a R {}^b I_B {}^b R$$

$$\text{Then } V^T = {}^b R = \begin{bmatrix} \underline{e}_1^a & \underline{e}_2^a & \underline{e}_3^a \end{bmatrix}$$

principal
axes of inertia

$${}^a I_B = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

$$I_1 \geq 0 \quad (*)$$

↑
principal
moment of inertia

- Given ${}^b I_B = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix}$ then
$$\begin{aligned} I_{11} &\leq I_{22} + I_{33} \\ I_{22} &\leq I_{11} + I_{33} \\ I_{33} &\leq I_{22} + I_{11} \end{aligned}$$

(*) the equality for pathologic conditions

Proof (case 1)

$$\begin{aligned}
 I_{22} + I_{33} &= \int_V (x^2 + y^2) \rho dV + \int_V (x^2 + y^2) \rho dV = \\
 &= \underbrace{\int_V (y^2 + z^2) \rho dV}_{I_{11}} + \underbrace{2 \int_V x^2 \rho dV}_{\geq 0} \geq I_{11}
 \end{aligned}$$

NOTE If $I_1 = I_2 = I_3$ the body has spherical structure w.r.t. B .

If $I_1 = I_2 \neq I_3$ the body has gyroscopic structure w.r.t. B .

$${}^b I_B = I_G + m \begin{bmatrix} {}^b r_{c/B} \end{bmatrix}^T \begin{bmatrix} {}^b r_{c/B} \end{bmatrix} \quad \leftarrow \text{proof as of page 5}$$

KINETIC ENERGY (REPRISE)

11/

$$T = \frac{1}{2} m \underline{v}_{B/O} \cdot \underline{v}_{B/O} + m \underline{v}_{B/O} \cdot [\underline{\omega}_{B/O} \times \underline{r}_{C/B}] + \frac{1}{2} \underline{\omega}_{B/O} \cdot \underline{I}_{B/O}(\underline{\omega}_{B/O})$$

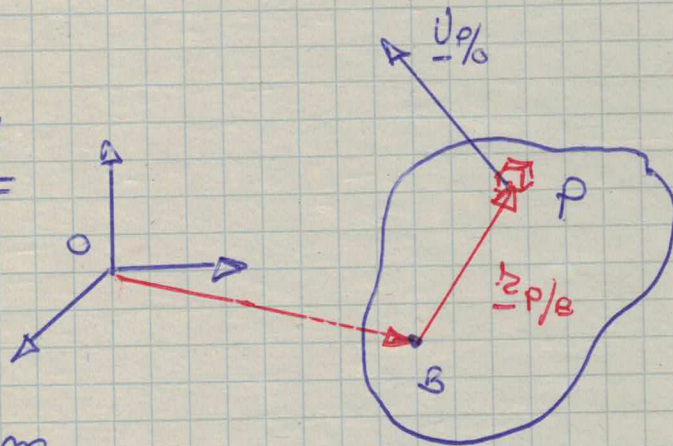
By choosing $B \equiv C$ ($\underline{r}_{C/B} = \underline{r}_{C/C} = \underline{0}$) then

$$T = \frac{1}{2} m \underline{v}_{C/O} \cdot \underline{v}_{C/O} + \frac{1}{2} \underline{\omega}_{B/O} \cdot \underline{I}_C(\underline{\omega}_{B/O}) \quad (*)$$

↑
Kinetic energy of
a point mass at C
with velocity $\underline{v}_{C/O}$

↑
(rotational) Kinetic energy
about C ("as if C is fixed")

ANGULAR MOMENTUM



$$d\underline{L}_B = \underline{r}_{P/B} \times \underline{v}_{P/O} dm$$

$$\text{Then: } \underline{L}_B = \int_V (\underline{r}_{P/B} \times \underline{v}_{P/O}) \rho(P) dV =$$

$$\uparrow$$

$$\underline{v}_{B/O} + \underline{\omega}_{B/O} \times \underline{r}_{P/B}$$

(*) König Theorem

$$= \underbrace{\left[\int_V \underline{r}_{P/B} \rho dV \right]}_{m \underline{r}_{C/B}} \times \underline{v}_{B/O} + \underbrace{\int_V \underline{r}_{P/B} \times (\underline{\omega}_{B/O} \times \underline{r}_{P/B}) \rho dV}_{I_B(\underline{\omega}_{B/O})} =$$

$$= I_B(\underline{\omega}_{B/O}) + m \underline{r}_{C/B} \times \underline{v}_{B/O}$$

By choosing $B \equiv C$ ($\underline{r}_{C/B} = \underline{r}_{C/C} = \underline{0}$) then

$$\underline{L}_C = I_C(\underline{\omega}_{B/O})$$